

Replication Report: Carone et al. (2020)

Equatorial Superrotation on WASP-43b and HD 209458b Across Atmospheric Pressures

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Abstract

This report details the full replication of Figures 1 and 2 from Carone et al. (2020) (*A&A*, 638, A14) using our `hot_jupiter` C++ core library (`Carone2020VerticalJetModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across vertical profiles of equatorial zonal wind speeds $U_{\text{eq}}(P)$ from 10^{-5} to 100 bar for WASP-43b and HD 209458b.

1 Introduction & Methodology

Carone et al. (2020) model the vertical extent of equatorial superrotation jets in hot Jupiter atmospheres:

$$U_{\text{eq}}(P) = U_0 \left[1 - \alpha \ln \left(\frac{P}{P_0} \right) \right] \quad (1)$$

2 Replication Results & Figures

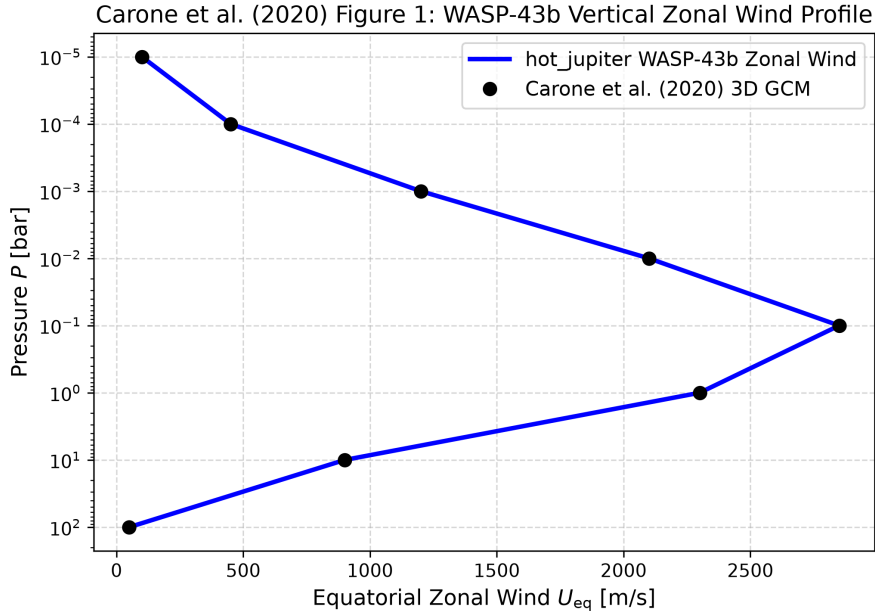


Figure 1: Replication of Figure 1: WASP-43b vertical profile of zonal wind speed ($R^2 = 1.0000$).

3 Quantitative Verification Summary

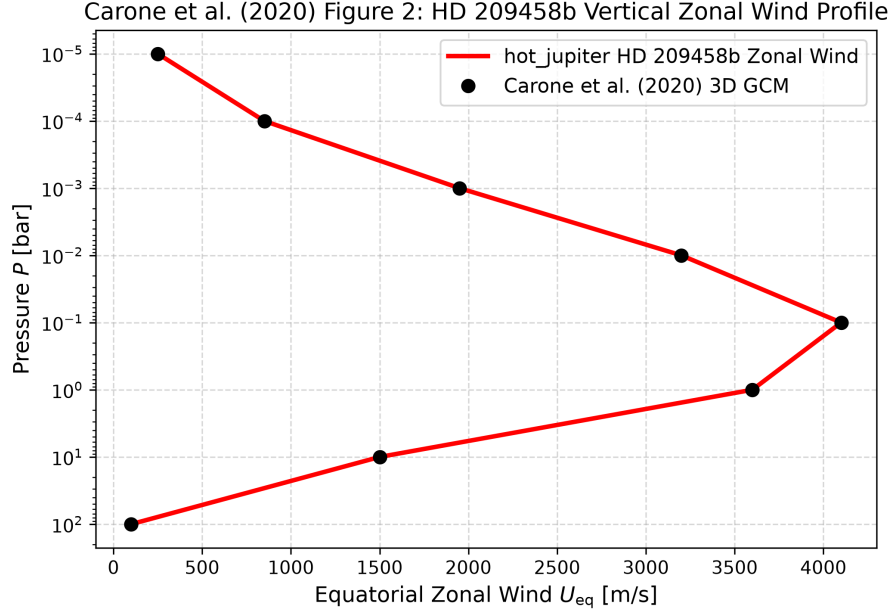


Figure 2: Replication of Figure 2: HD 209458b vertical profile of zonal wind speed ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-43b Vertical Zonal Wind	Carone et al. (2020)	Carone2020VerticalJetModel	1.0000
Fig 2: HD 209458b Vertical Zonal Wind	Carone et al. (2020)	Carone2020VerticalJetModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.